

A BRIEF HISTORY OF EVERYONE WHO EVER LIVED

The Human Story Retold
Through Our Genes

ADAM RUTHERFORD

FOREWORD BY SIDDHARTHA MUKHERJEE



A BRIEF HISTORY OF EVERYONE WHO EVER LIVED: *The Human Story Retold
Through Our Genes*

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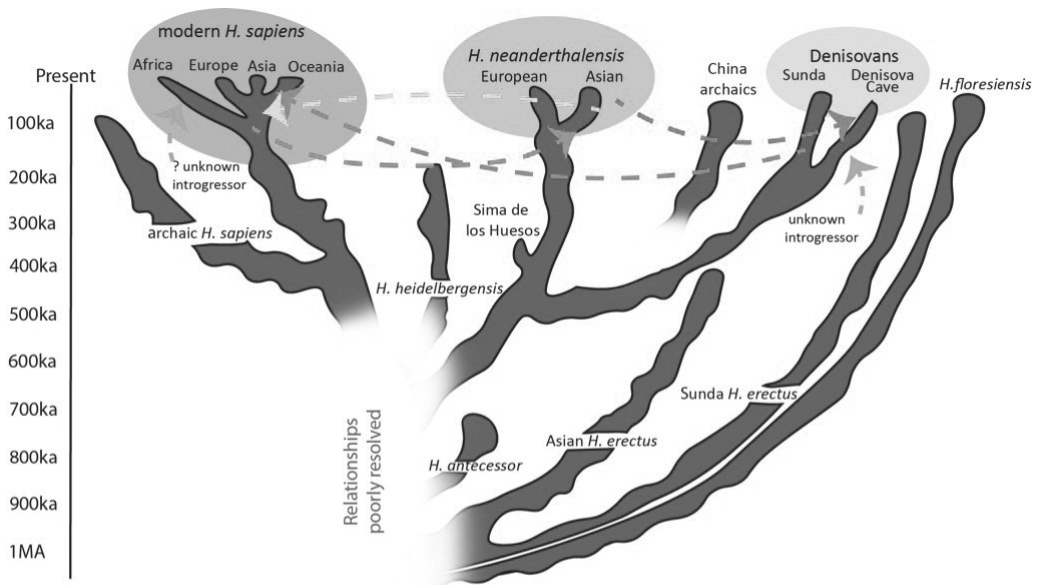
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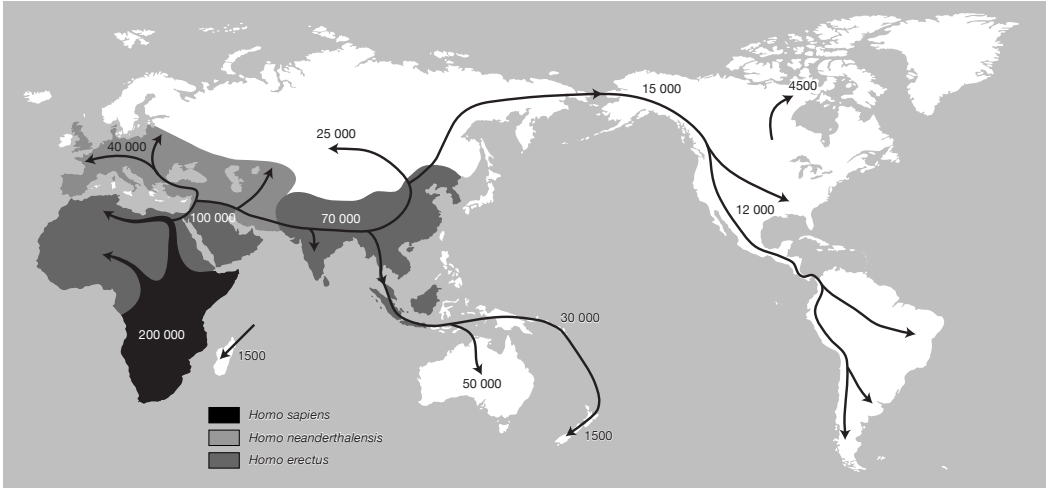
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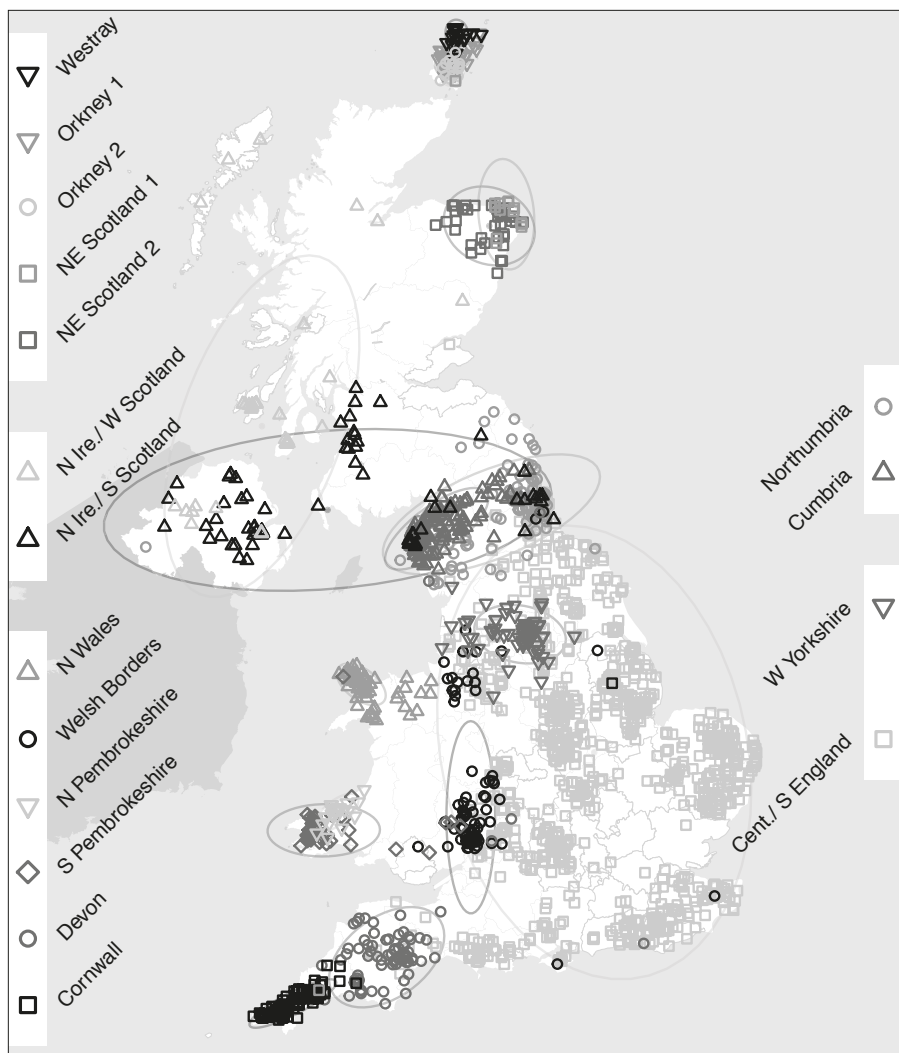
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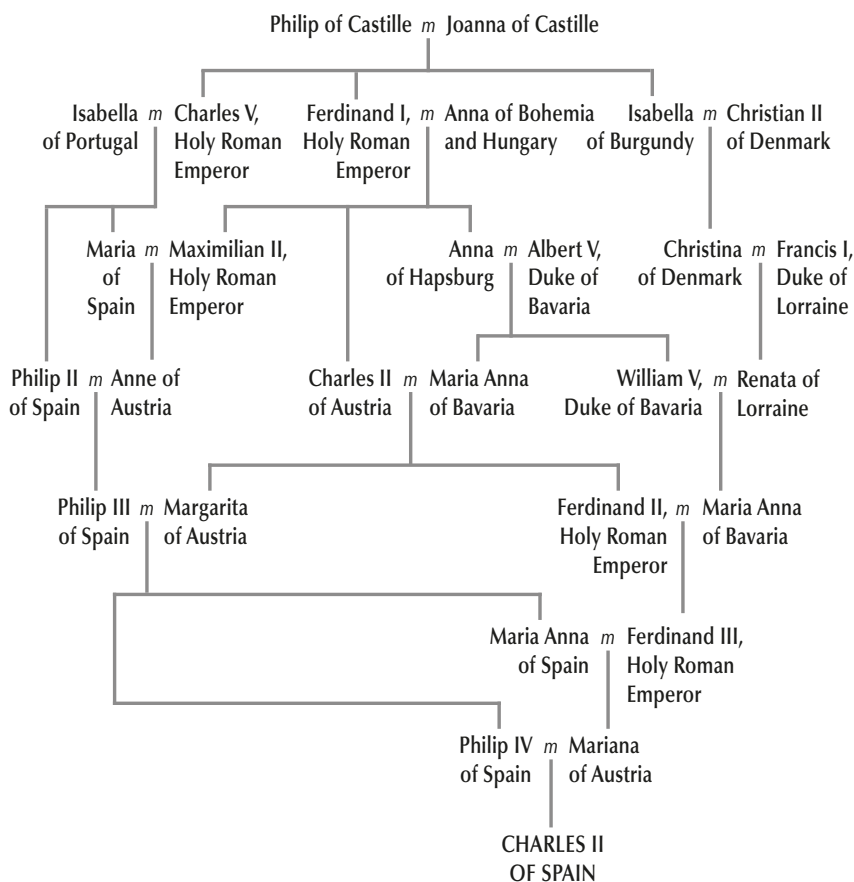
The murky evolutionary shrub of humankind. *Old bones, combined with new analysis of old DNA, has meant that what was once a confident branching tree has been pollarded and pruned and replanted as an unrooted bush. The broad blobs represent individual human species, and the dotted lines are the flow of genes via sex between them. The more we learn, the messier the picture becomes.*



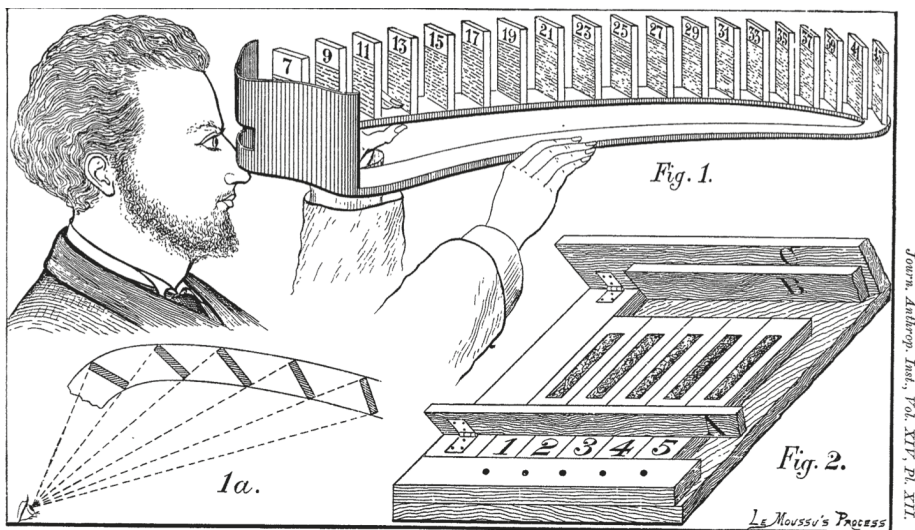
The migration of *Homo sapiens* out of Africa. Anatomically modern humans began their tenure on Earth in eastern Africa, around 200,000 years ago. They had begun to trickle out of Africa at least 100,000 years ago. They met Neanderthals in Europe, and other human species en route, and according to our DNA, bred with many of them.



The fine-scale genetic map of the people of Great Britain. *The People of the British Isles project sorted the DNA from 2,039 Britons into genetically similar clusters, and placed them on a map according to where they were born. The map reveals local constellations that reflect the history of the British Isles. (© Stephen Leslie, University of Melbourne, Australia. Based on data published in Nature 2015. Contains OS data © Crown copyright and database right 2012. © EuroGeographics for some administrative boundaries.)*



The family tree of Charles II of Spain. *The Hapsburg family stopped outbreeding more than a century before Charles II's birth in 1661. Six generations should contain 62 different ancestors; Charles II's has 32. Eight generations should contain 254 different ancestors; Charles II's has 82. This is what we might refer to as 'suboptimal'.*



ANTHROPOMETRIC INSTRUMENTS.

Journ. Anthropol. Inst., Vol. XIV, Pl. XII.

Galton's biometric equipment. *Figure 1 shows a sickle-shaped layout for testing acuity of vision, first with one eye and then the other. The pages are from the Shilling Diamond edition of the Prayer Book (though he points out that familiar texts are 'objectionable' to the test, and – ever the mathematician – 'a page of logarithms would be better'). Figure 2 depicts the 'colour sense' test, in which the participant is asked to identify shades of green wool in the five lengthways cartridges.*

GENE SWEEPSTAKE 2000-2003

in <http://www.gensemb1.org/genesweep.html>

① costs \$1 2000, \$5 2001, \$3 2002 and ~~\$4 2003~~ to bet.

② Bets are for one number, winner takes all

③ A gene is a set of connected exons by transcription / mRNA splicing + protein coding ^{X.B.v.*†}

④ Assessment of Gene number ^{method} will occur via agreement on 2002 CSHL meeting

⑤ Actual assessment of Gene number will occur on the 2003 CSHL Genome meeting (or equivalent)

⑥ Write your email, number, + payment in the book

⑦ One bet per person, per year

⑧ No pencil bets

⑨ Stay at CSHL. Contact David Stewart

~~Stewart~~ Stewart@cschl.org

α cds in repetitive regions are not counted even if expressed.

γ Autosomal + X + Y chromosomes from the reference sequence. ^{no mitochondrial genome.}

* At least one transcript must encode a protein.

β If ~ 1 cell gene are only one gene per loci

+ 7 transcribing exons, each trans splice is separate gene

The gene sweepstake betting book. Ewan Birney opened a betting book at a meeting of the Human Genome Project in 2000. For a dollar a bet, he asked the world's top geneticists to predict how many genes a human has. The prize for the nearest guess was the pot and a bottle of Scotch. The winning bet, by Lee Rowen, was 25,947. The real number is around 20,000.

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Alice Roberts helped me more than could reasonably be expected, I think in atonement for the many times she has

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Glossary

allele

Although we all carry the same set of genes, they are not identical in all of us. An allele is a variant of a gene (or position in the genome) akin to an alternative spelling, or a typo. The US version of this book will contain the word 'behavior', whereas in the UK it is 'behaviour', which doesn't change its meaning, whereas the single letter change in 'affect' and 'effect' alters meaning from 'to change' to 'result'. Both are examples of alleles.

amino acid

Small molecules that link together to form proteins. Each amino acid is encoded in DNA by a particular arrangement of three bases.

bases

Also called nucleotides, or referred to as letters; the individual components of DNA, the arrangement of which determines the genetic code. There are four bases, abbreviated to A, T, C and G. The double helix in a sense is like a twisted ladder, the rungs of which are made of pairs of bases; A only ever pairs with T, C only ever with G. (In RNA, the T is replaced with a U.)

BCE

Before the Common Era. Synonymous with the term

‘Before Christ’, but science has replaced it, appropriately, with something universal.

CE

The Common Era, synonymous with the old fashioned Anno Domini, i.e. the year this book was first published was 2016 CE.

chromosome

Long stretches of DNA that harbour genes. Species have a specific number of chromosomes. In humans it’s twenty-two pairs of autosomes, plus two sex chromosomes: two Xs if you are a woman, and an X and a Y if you are not.

codon

Three bases in a particular order encode a specific amino acid, which link together to form proteins. There are four bases, therefore sixty-four possible combinations if arranged into threes. But biology only uses twenty amino acids (plus a signal to end a protein, STOP), which means that the genetic code has redundancy in it: several different codons can encode the same amino acid. A stretch of codons in DNA marks out a gene.

DNA

Deoxyribonucleic acid: the material of genetics, typically arranged into chromosomes. DNA is a script in which genes are written.

gene

In its simplest form, a gene is a sequence of DNA that encodes a working protein. This definition adheres to what

is known as the ‘central dogma’: DNA makes RNA makes protein. Genes are composed of DNA, and these lie as part of chromosomes, like sentences in text. There is no more precise definition, but there are active genetic elements that do not follow this model. In the last few years, many bits of DNA that encode RNA which never becomes a protein have been identified. These may yet be classified as genes.

genetics

The study of genes, DNA, disease, inheritance, evolution and a whole lot more.

genome

The entire genetic material of an organism, the sum total of its DNA. The Human Genome Project was a huge publicly funded collaborative scientific endeavour that concluded in the first few years of the twenty-first century. Its aim – which was delivered under budget and on time – was to provide a complete readout of a human genome as a database which we could go on to mine for information about how humans work, how we evolved and what happens when we go wrong. The principal output was a reference genome, an average human from which we can compare our infinite variation.

genomics

The study of genomes, including genetics but much more as well. In studying genomes we look at more than simply the genes that have specific functions in an organism, and analyse the regulation of those genes, and the information stored in DNA.

genotype

Genotype refers to the version of the genes that you possess.

heterozygous/homozygous

These refer to the pair of each gene you have, one inherited from each parent. Heterozygous means you have two different versions of the same gene; homozygous means they are both the same.

Mendelian/Mendel's Laws

The basic rules of biological inheritance were set by Gregor Mendel in the nineteenth century, as a result of his experiments in which he bred thousands of pea plants and observed how various characteristics were passed from generation to generation. There are three broad rules, which can be summarised thus:

1. Organisms have two copies of each allele, one inherited from each parent.
2. Each trait is inherited independently of any other.
3. Some alleles are dominant over others, which are called recessive. If you inherit one dominant from one parent, and one recessive from the other, only the dominant will be expressed. To see a recessive trait, you must have inherited two recessive alleles, one from each parent, for example with the alleles for red hair.

As is so often the case, these laws are more like rules, in that there are some notable exceptions. Biology is annoying like that.

mitochondria

Tiny compartments within cells that are primarily responsible for generating energy for that cell. They contain their own chromosome, a small loop of DNA separate from the vast majority contained within the nucleus of a cell. Mitochondrial DNA (which is sometimes written as mtDNA) is of great interest to geneticists and genealogists as it is only ever passed on from mother to child, and thus can exclusively chart a matrilineage.

phenotype

The phenotype is the physical manifestation of a gene or genotype. For example, a particular version of the gene MC1R (the genotype) will mean your phenotype will be a redhead.

protein

The primary functional biological molecules in living things. They are made up of simple molecules called amino acids assembled into long strings. These fold up to make a three-dimensional structure, and often assemble with other proteins in cells to enact their function. All life is made of, or by, proteins.

single nucleotide polymorphism

Many of the genetic differences between people are made up of individual letters of genetic code being different at specific points in the genome (see **allele**, above). When these are simply alterations in a single letter at a specific point, we call them single nucleotide polymorphisms, which is universally abbreviated to SNP (and pronounced 'snip'). The words 'inquiry' and 'enquiry' would be an alphabetical equivalent of a SNP.

References and further reading

I urge you to read the following books that have influenced me over the years on the subject of genetics, evolution and humans.

The Language of the Genes: Solving the Mysteries of Our Genetic Past, Present and Future by Steve Jones (HarperCollins, 1991) is a classic, written well before the Human Genome Project was begun and so twenty-five years later is out of date. However, the stories within and Jones's peerless storytelling make it well worth reading to this day.

Genome: A Biography in 23 Chapters by Matt Ridley (Fourth Estate, 1999): also a classic, and also published before the human genome was unveiled.

Life's Greatest Secret: The Race to Crack the Genetic Code by Matthew Cobb (Profile, 2015) is the most definitive version of the story of the genetic code that I have read.

And of course, always return to the source:

The Descent of Man, and Selection in Relation to Sex by Charles Darwin, MA, FRS (John Murray, 1871)

Introduction

Estimate of the total number of humans born in the last 50,000 years

<http://www.prb.org/Publications/Articles/2002/HowManyPeopleHaveEverLivedonEarth.aspx>

‘There is no-one alive more youer than you!’ from *Happy Birthday to you!* by Dr Seuss (HarperCollins Children’s Books, 2005)

‘They fuck you up . . .’ excerpt from Larkin, Philip, ‘This be the Verse’, *High Windows* (Faber and Faber, 1974). Reproduced with permission from Faber and Faber Ltd

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Chapter 1: Horny and mobile

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